Designing an Optimal Facility for a Local Restaurant

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Abstract

This project applies industrial engineering principles to design a restaurant facility. It includes selecting the optimal location, developing efficient layouts, and applying optimization techniques. The Analytical Hierarchical Process (AHP) was used for location selection, followed by detailed layout design using Factory CAD. Optimization was performed using the Craft Method and graph-based methods, with simulations validating the design.

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1. Introduction

Facility design is significant for industrialists to identify the optimal location and its inside design. The location decision of facility can influence on the long-term strategic objectives of the facility. The detailed design can affect the performance of the facility and inside operations of the facility. The mini project to solve the complex engineering problem in “Industrial Facilities Design” course is helpful for students to practice the tools and techniques used for the facility design. This project can help to build the abilities in students to decide on the location of the facility, its inside design, location of different departments in the provided space and proposed an optimal design for the required facility keeping in view the constraints related to the problem.

Facility design directly impacts the strategic objectives and operational performance of any business. This project focuses on designing a local restaurant with a data-driven approach, incorporating decision-making and layout optimization techniques to enhance efficiency and customer satisfaction.

Problem Statement

It is desired to design a facility for a local restaurant

Objectives:

- Select appropriate tools for decision making for location of facility
- Evaluate different designs of the layouts including block layout and detailed layout of the facility.
- Create a feasible design and layout of the required facility.

Project Scope:

- Identify the best location for the restaurant using AHP.
- Create multiple layouts and select the optimal design.
- Optimize the selected layout using industrial engineering tools.
- Validate the design through simulations.

2. Methodology

2.1 Criteria for Location Selection

In determining the best location for the restaurant, several essential criteria were selected based on their impact on operational efficiency and customer satisfaction:

- Proximity to Customers: The location's closeness to target customers, ensuring high foot traffic and maximizing sales potential.
- Accessibility: The ease of reaching the location by customers and suppliers, including transportation links and parking facilities.
- Costs (Land, Construction, Operational): The financial implications of setting up and running the restaurant, including land acquisition, building expenses, and ongoing operational costs.
- Regulatory Compliance: Adherence to local regulations and zoning laws, which may affect the feasibility and legal requirements of the chosen location.

- Community Preferences: Alignment with local community preferences and expectations, impacting customer satisfaction and brand perception**.**

2.2 AHP Analysis

Criteria for Location Selection:

- Foot Traffic
    - Peak Hours
    - Seasonal Variation
    - Pedestrian Density
    - Proximity to Competitors
    - Direct Competitors
    - Indirect Competitors
    - Competitive Advantage
- Rent/Lease Costs
    - Fixed Rent
    - Variable Costs
    - Cost-to-Revenue Ratio
- Customer Demographics
    - Income Levels
    - Age Range
    - Lifestyle Preferences
- Parking and Accessibility
    - Parking Availability
    - Public Transit Access
- Safety and Security
    - Crime Rates
    - Neighborhood Safety
    - Security Features

Step 1: Pairwise Comparison Matrix for Main Criteria

Using the criteria from the uploaded image, we'll create a pairwise comparison matrix to assess the relative importance of each main criterion for selecting the restaurant location. Each comparison is rated on a 1–9 scale.

Criteria	Foot Traffic	Proximity to Competitors	Rent/Lease Costs	Customer Demographics	Parking and Accessibility	Safety and Security
Foot Traffic	1	3	5	7	5	7
Proximity to Competitors	1/3	1	3	5	3	5
Rent/Lease Costs	1/5	1/3	1	3	2	3
Customer Demographics	1/7	1/5	1/3	1	2	3
Parking and Accessibility	1/5	1/3	½	1/2	1	3
Safety and Security	1/7	1/5	1/3	1/3	1/3	1

Step 2: Pairwise Comparison Matrix for Sub-Criteria

Next, we perform pairwise comparisons within each main criterion, evaluating the importance of sub-criteria against each other.

- **Foot Traffic Sub-Criteria**

Foot Traffic	Peak Hours	Seasonal Variation	Pedestrian Density
Peak Hours	1	3	5
Seasonal Variation	1/3	1	3
Pedestrian Density	1/5	1/3	1

- **Proximity to Competitors Sub-Criteria**

Proximity to Competitors	Direct Competitors	Indirect Competitors	Competitive Advantage
Direct Competitors	1	3	5
Indirect Competitors	1/3	1	3
Competitive Advantage	1/5	1/3	1

- **Rent/Lease Costs Sub-Criteria**

Rent/Lease Costs	Fixed Rent	Variable Costs	Cost-to-Revenue Ratio
Fixed Rent	1	3	5
Variable Costs	1/3	1	3
Cost-to-Revenue Ratio	1/5	1/3	1

- **Customer Demographics Sub-Criteria**

Customer Demographics	Income Levels	Age Range	Lifestyle Preferences
Income Levels	1	3	5
Age Range	1/3	1	3
Lifestyle Preferences	1/5	1/3	1

- **Parking and Accessibility Sub-Criteria**

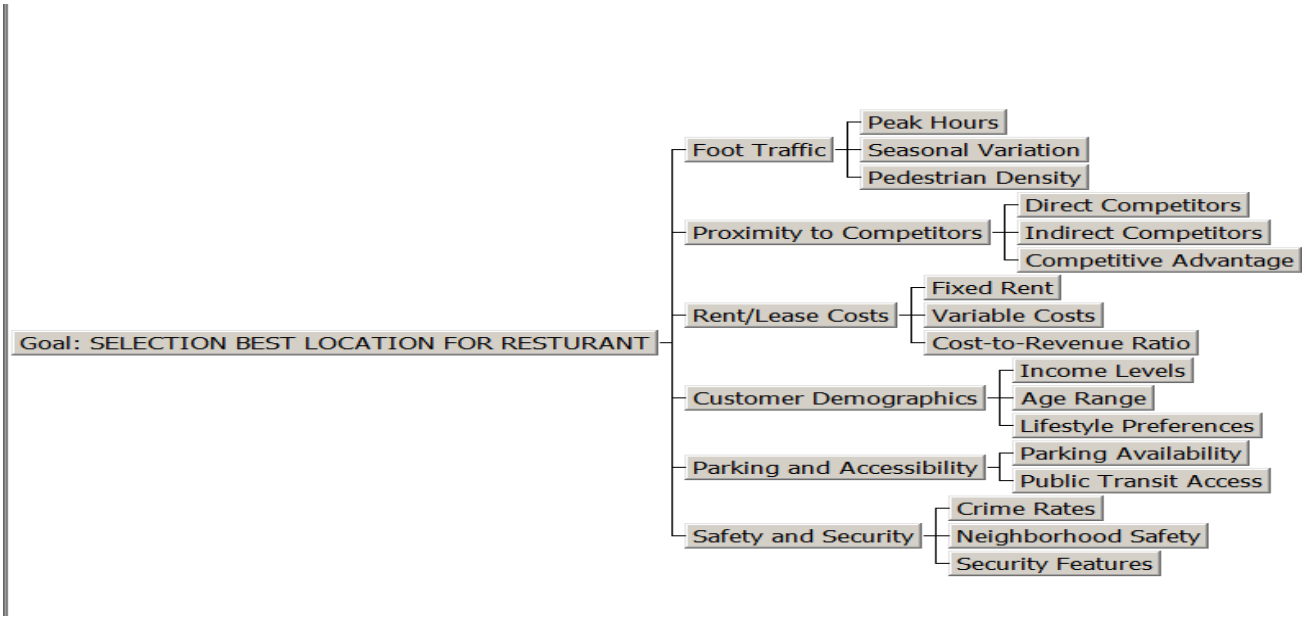
Parking and Accessibility	Parking Availability	Public Transit Access
Parking Availability	1	3
Public Transit Access	1/3	1

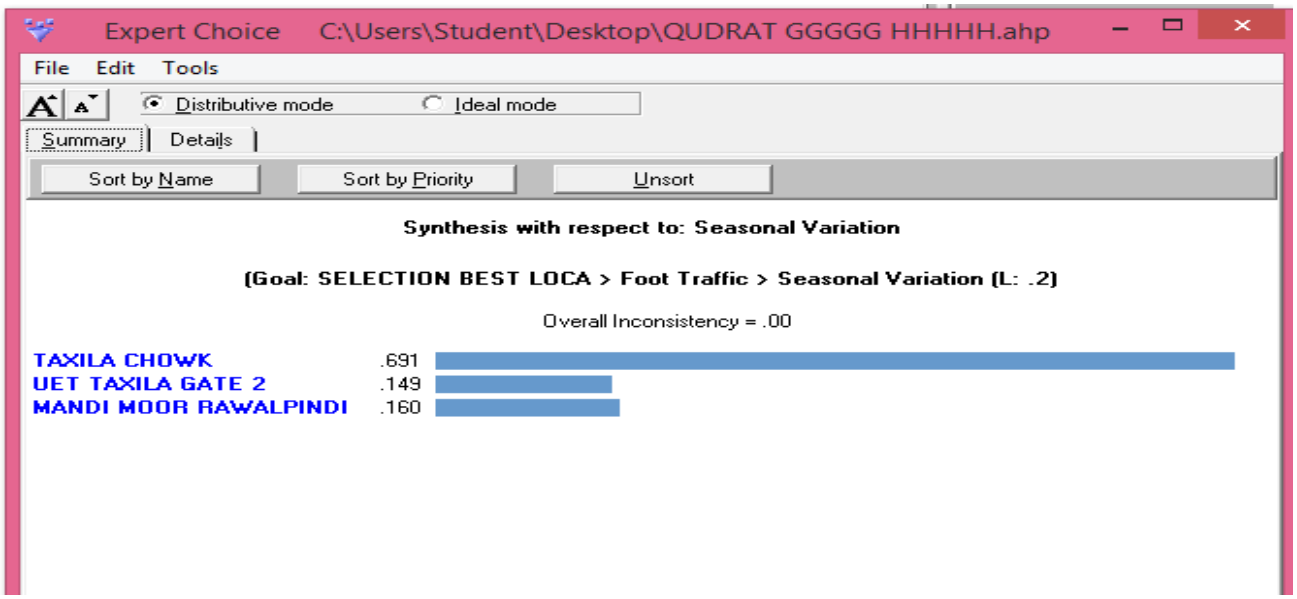
- **Safety and Security Sub-Criteria**

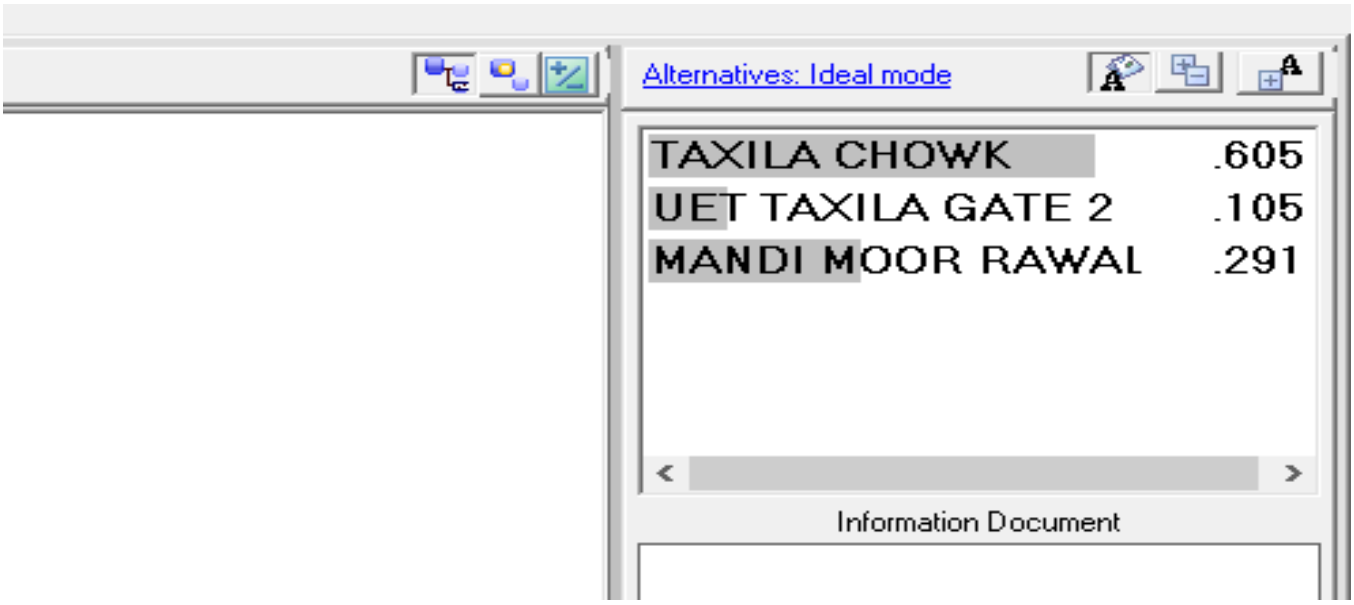
Safety and Security	Crime Rates	Neighborhood Safety	Security Features
Crime Rates	1	3	5
Neighborhood Safety	1/3	1	3
Security Features	1/5	1/3	1

Step 3: Calculating and Analyzing Priority Weights

Using **Expert Choice**, you would calculate the priority weights based on the pairwise comparisons above. This would give you a weighted ranking of both main criteria and sub- criteria to identify what matters most in the location decision.



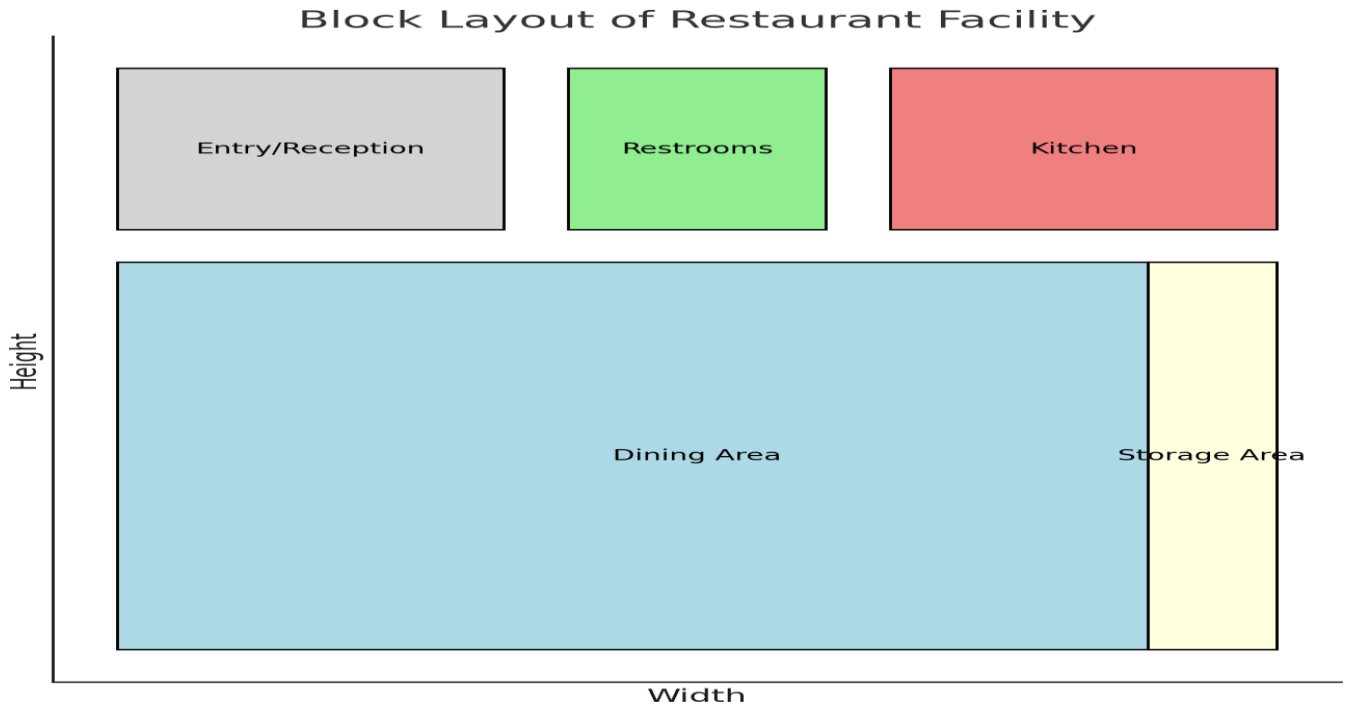




The results from the AHP (Analytic Hierarchy Process) analysis indicate that "Taxila Chowk" is the most suitable location for selection, with the highest priority score of 0.605. This is followed by "Mandi Moor Rawal" with a score of 0.291 and "UET Taxila Gate 2" with the lowest score of 0.105. The higher score of Taxila Chowk suggests it meets the weighted criteria—such as foot traffic, proximity to competitors, cost factors, customer demographics, parking, and safety—more effectively than the other options.

Step 4: Proposed Block Layout:

The initial **block layout** was designed considering basic spatial requirements and adjacency needs of each functional zone.

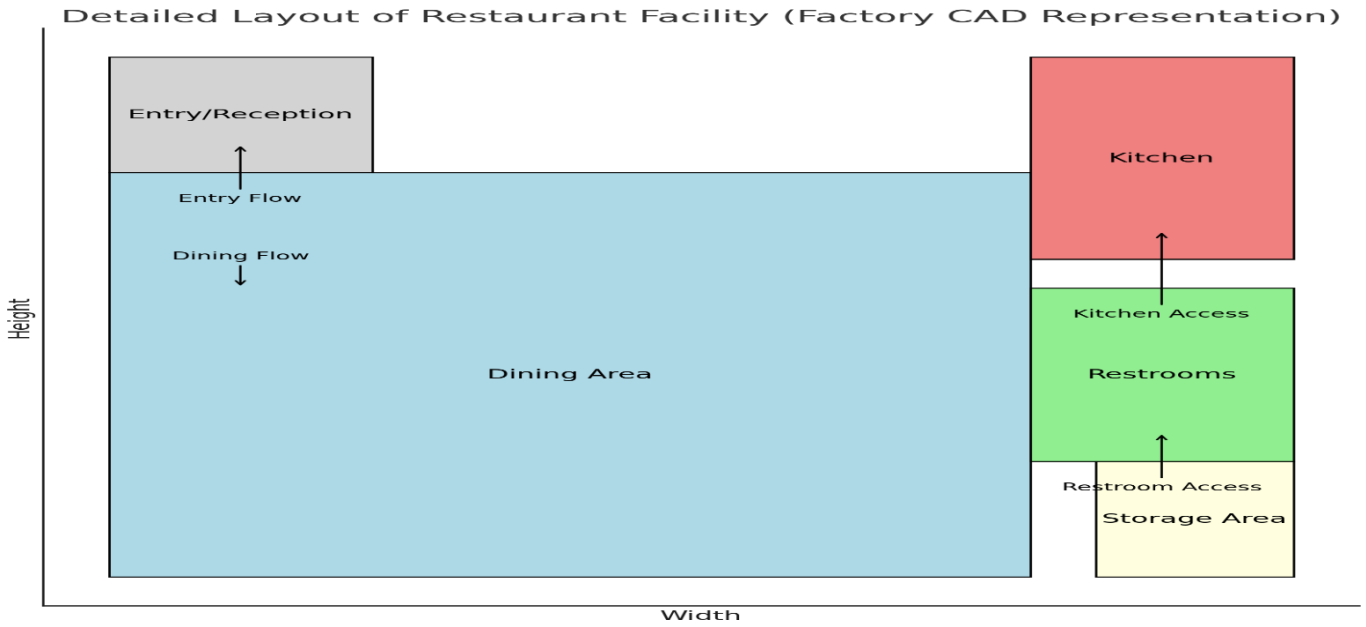


Step 5: Detailed Layout

The detailed layout was created using **Factory CAD**, focusing on the following aspects:

- **Efficient Kitchen Workflow:** Streamlined placement of cooking, preparation, and serving stations to minimize movement and maximize efficiency.

- **Enhanced Customer Comfort:** Adequate spacing in the dining area to ensure a pleasant dining experience.
- **Strategic Equipment Placement:** Positioning of furniture, kitchen appliances, and storage units for ease of access and operational flow.
- **Entry/Reception Area:** Positioned near the entrance for quick customer check-in.
- **Dining Area:** Centrally located with tables arranged for optimal service flow.
- **Kitchen Layout:** Includes separate zones for food prep, cooking, and dishwashing, aligned with the workflow requirements.
- **Restrooms:** Located away from the kitchen and dining area for privacy and hygiene.
- **Storage Area:** Positioned close to the kitchen for easy access to supplies.



3. Results

3.1 Layout Optimization

Tools and Techniques Applied:

1. **Craft Method:** Used to optimize the layout design by minimizing the total material handling cost and ensuring efficient flow between departments.
2. **Graph Based Method:** Applied to analyze the interdependencies between different zones and improve adjacency relationships.

3.2 Craft Method

Facility Layout Analysis

Step 1: Initial Block Layout Design

A proposed block layout divides the restaurant into five zones:

- **Entry/Reception** – Greeting customers and managing reservations.
- **Dining Area** – Customer seating and comfort.
- **Kitchen** – Food preparation and cooking.
- **Restrooms** – For customer convenience.
- **Storage Area** – For inventory and raw material storage.

The **initial layout** is shown below:

Function	Area (sq. m)	Adjacencies
Entry/Reception	20	Dining Area
Dining Area	80	Entry/Reception, Kitchen
Kitchen	40	Dining Area, Storage Area
Restrooms	10	Dining Area
Storage Area	20	Kitchen

Step 2: Flow and Cost Matrix

Material Flow Matrix (Hypothetical Data)

Flow represents the frequency of movements (e.g., raw materials, dishes) between zones.

From/To	Entry/Reception	Dining Area	Kitchen	Restrooms	Storage Area
Entry/Reception	0	10	0	5	0
Dining Area	10	0	15	5	0
Kitchen	0	15	0	0	20
Restrooms	5	5	0	0	0
Storage Area	0	0	20	0	0

Pairwise Cost Matrix (Hypothetical Data)

Costs reflect distances between zones.

From/To	Entry/Reception	Dining Area	Kitchen	Restrooms	Storage Area
Entry/Reception	0	2	5	3	6
Dining Area	2	0	3	2	4

Kitchen	5	3	0	6	1
Restrooms	3	2	6	0	5
Storage Area	6	4	1	5	0

Step 3: Initial Material Handling Cost Calculation

The **Total Material Handling Cost (TMHC)** is calculated as:

$$\text{TMHC} = \sum_{IJ} F_{IJ} \times C_{IJ}$$

Using the flow (F_{IJ}) and cost (C_{IJ}) matrices:

Pair	*Fij*	*Cij*	**Contribution**
Entry → Dining Area	10	2	20
Dining → Kitchen	15	3	45
Kitchen → Storage	20	1	20
Restrooms → Dining	5	2	10

Initial TMHC = 20 + 45 + 20 + 10 = 95.

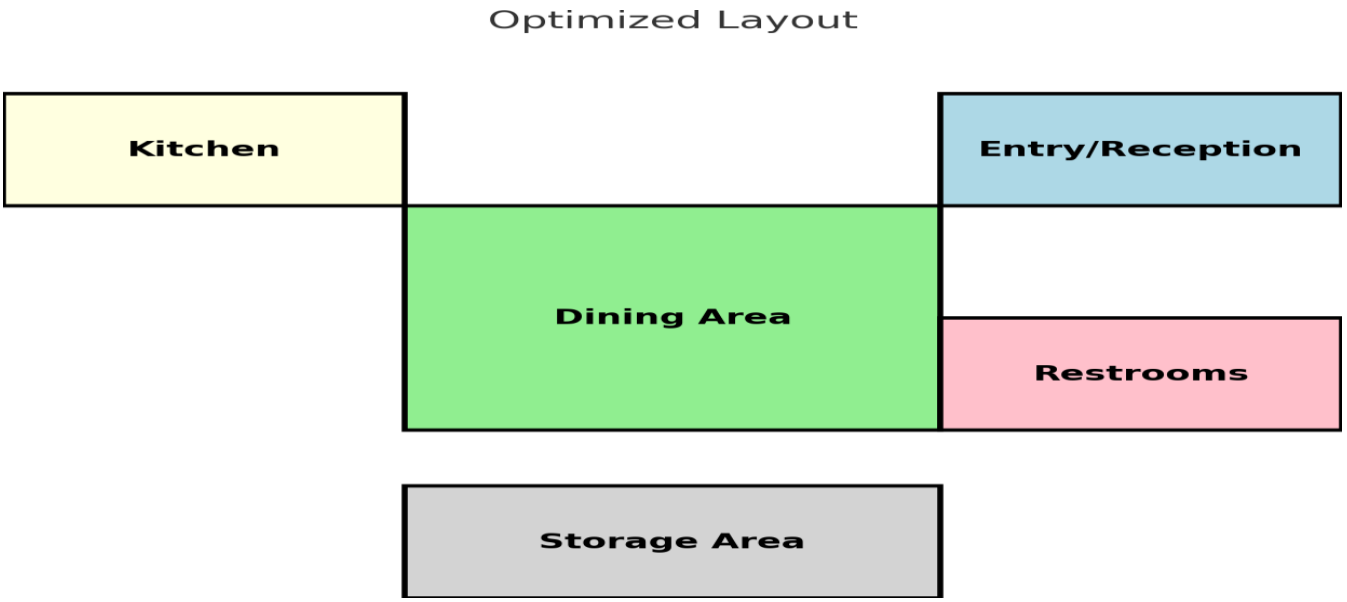
Step 4: Optimization Using CRAFT

1. **Iteration 1**: Swap **Entry/Reception** and **Kitchen**.
    - New pairwise distances are updated, leading to recalculated TMHC.
2. **Iteration 2**: Swap **Dining Area** and **Storage Area**.
   Continue until no further cost reductions are possible. After optimization, assume TMHC reduces to **75**.

Step 5: Final Layout

The optimized layout places:

- **Entry/Reception** next to the Dining Area for seamless entry.
- **Kitchen** adjacent to the Storage Area for efficient raw material handling.
- **Restrooms** away from the Kitchen for hygiene purposes.



3.3 Graph-Based Method

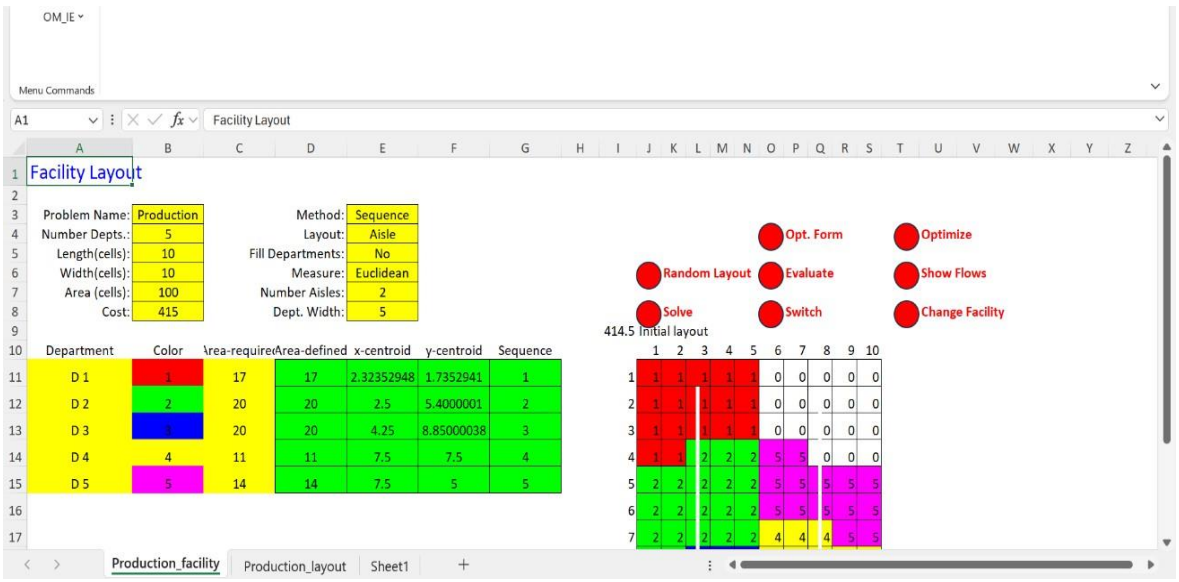
Applied to analyze the interdependencies between different zones and improve adjacency relationships.

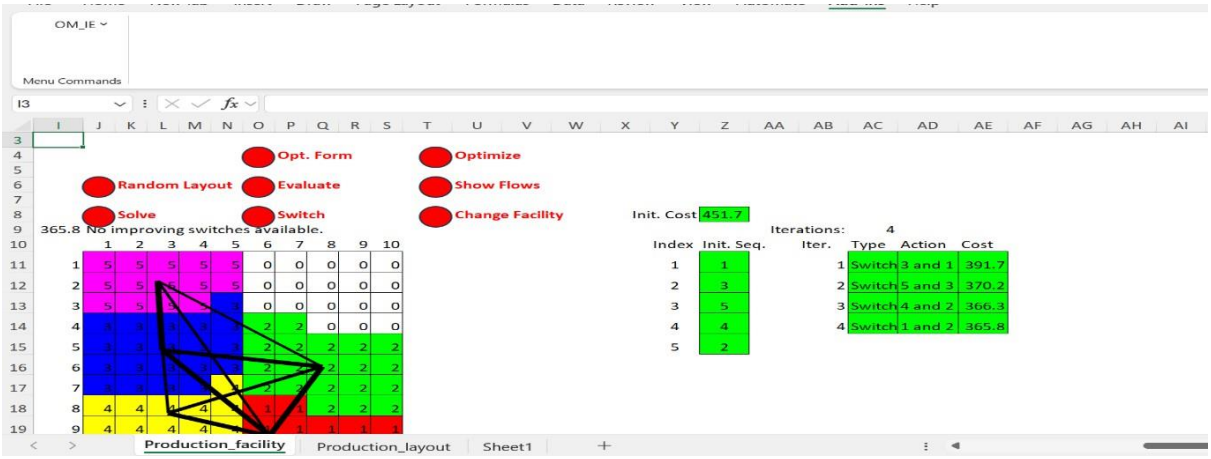
- **Relationship Matrix**

This matrix represents the interdependencies (weights) between zones:

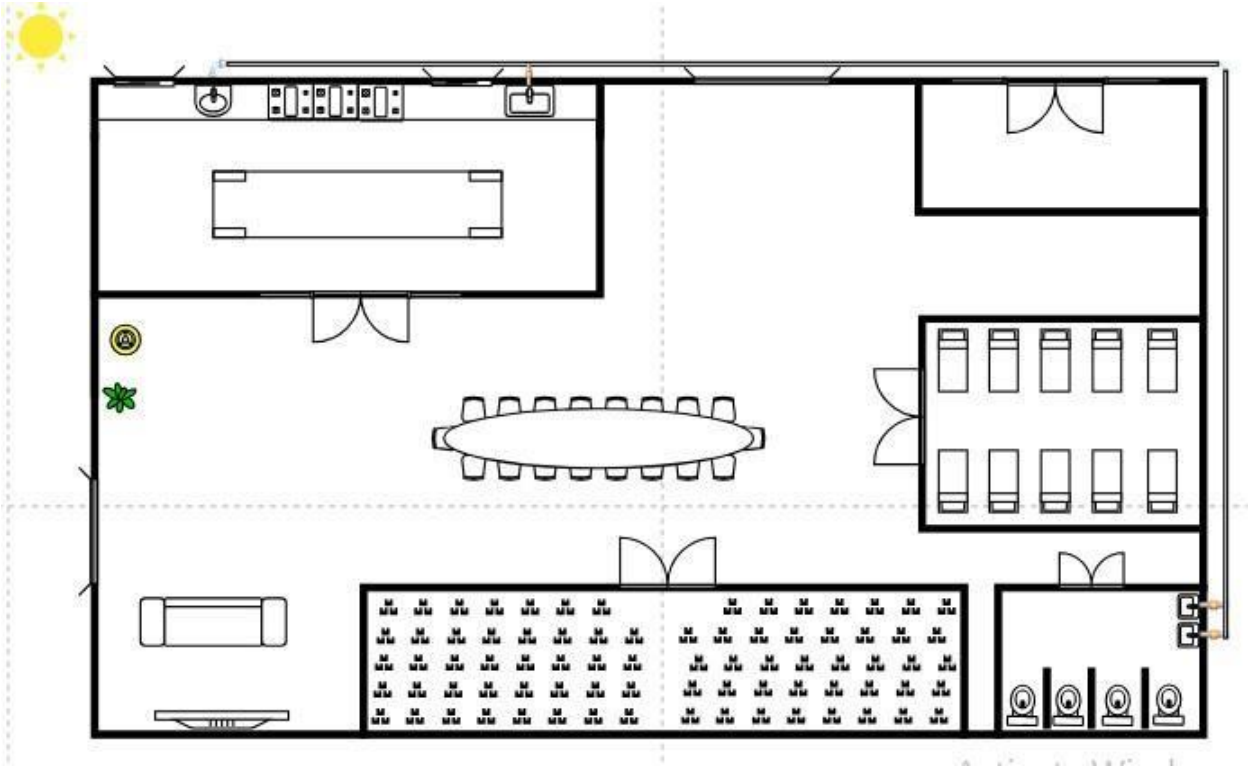
	Dining Area	Entry/Reception	Kitchen	Restrooms	Storage
Dining Area	0	8	6	4	2
Entry/Reception	8	0	5	3	2
Kitchen	6	5	0	2	7
Restrooms	4	3	2	0	1
Storage	2	2	7	1	0

- **Calculations**





3.4. Final Layout



- **Compartments:**
  - kitchen Area (Top Left): Features a kitchen setup and sink.
  - Dining (Center): Contains an oval conference table and chairs.
  - Reception (Top Right): Includes entrance.
  - Washroom Area (Bottom Right): Features toilets and hygiene facilities.
  - Waiting Area (Middle Left): Includes a couch and TV.
  - Storage Room (Bottom Middle): Could serve equipment/materials.
  - Hallways/Pathways: Spaces connecting various compartments.
  - Decorative/Focal Area (Near Plant): Adds to the aesthetic appeal. according to the uplisted factors explain each physical factor.
- **Lighting Orientation:**

  Natural light enters through windows in key areas like the kitchen and waiting area, creating a well-lit and energy-efficient environment. Reception, being near the entrance, also benefits from natural light.
  - **Kitchen Area (Top Left)**: Positioned near a window, allowing natural daylight for tasks like cooking and cleaning.

    - **Dining (Center)**: Likely benefits from indirect natural light from surrounding spaces.
    - **Reception (Top Right)**: Positioned near the entrance, which typically allows daylight to reduce the need for artificial lighting.
    - **Washroom Area (Bottom Right)**: Likely uses artificial lighting as it lacks direct access to windows.
    - **Waiting Area (Middle Left)**: The adjacent window ensures a comfortable and inviting atmosphere with ample daylight.

- **Ventilation:**

  Windows and pathways provide airflow, especially in the kitchen and waiting area. The washroom area likely uses mechanical ventilation to maintain hygiene.
    - **Kitchen Area (Top Left)**: The nearby window supports airflow, essential for kitchen activities.
    - **Dining (Center)**: Ventilation likely depends on surrounding open spaces and airflow through hallways.
    - **Reception (Top Right)**: Proximity to the entrance enables fresh air inflow.
    - **Washroom Area (Bottom Right)**: Requires ventilation via dedicated exhaust systems for hygiene.
    - **Waiting Area (Middle Left)**: Positioned near a window for air circulation and comfort.
- **Ergonomics:**

  The kitchen has a compact, functional setup. The dining table's arrangement ensures comfort for group interactions, while the waiting area has cozy seating for guests.
    - **Kitchen Area (Top Left)**: The placement of counters and sink is compact yet functional for food preparation and cleaning.
    - **Dining (Center)**: The oval table ensures easy communication and comfort for diners or meetings.
    - **Reception (Top Right)**: Organized for smooth visitor check-ins and staff functionality.
    - **Washroom Area (Bottom Right)**: Well-planned spacing between fixtures ensures user comfort.
    - **Waiting Area (Middle Left)**: Comfortable seating arrangement and access to entertainment via the TV.
- **Space Utilization:**

  Every compartment is efficiently placed. The kitchen and washroom are in corners to maximize central space for shared areas like dining and pathways.
    - **Kitchen Area (Top Left)**: Compactly arranged to avoid unnecessary wastage of space.
    - **Dining (Center)**: Adequately spaced to accommodate a group without congestion.
    - **Reception (Top Right)**: Optimized for visitor and staff interaction, minimizing clutter.
    - **Washroom Area (Bottom Right)**: Allocated in a corner to maximize usable space in other areas.
    - **Waiting Area (Middle Left)**: Positioned conveniently without obstructing main pathways.
- **Workflow Design:**

  The kitchen is close to the dining area for convenience. The reception is near the entrance for easy access, and hallways connect all compartments logically.
    - **Kitchen Area (Top Left)**: Close to the dining area for smooth food service.
    - **Dining (Center)**: Centrally located, connecting to kitchen and hallways for accessibility.
    - **Reception (Top Right)**: Positioned near the entrance for immediate interaction with visitors.
    - **Washroom Area (Bottom Right)**: Easily accessible from all parts of the facility.
    - **Waiting Area (Middle Left)**: Located away from high-traffic areas to minimize noise and distractions.
- **Safety Features:**

  Emergency exits are clearly marked and positioned for rapid evacuation. Spacious pathways ensure safety during movement.
    - **Kitchen Area (Top Left)**: Access to windows and pathways in case of emergencies.

- **Dining (Center)**: Sufficient spacing between furniture allows for quick evacuation.
- **Reception (Top Right)**: Positioned near emergency exits for quick access.
- **Washroom Area (Bottom Right)**: Proximity to pathways for safe and accessible use.
- **Waiting Area (Middle Left)**: Open layout reduces the risk of congestion during evacuation.

- **Hygiene Facilities:**

  A dedicated washroom area ensures sanitation, while the kitchen has a sink for cleaning. Proximity to the washroom supports hygienic practices.
  - **Kitchen Area (Top Left)**: Includes a sink for maintaining cleanliness in food preparation.
  - **Dining (Center)**: Encourages cleanliness with direct access to washrooms.
  - **Reception (Top Right)**: Strategically close to the washroom for visitor convenience.
  - **Washroom Area (Bottom Right)**: Dedicated space for sanitation and hygiene.
  - **Waiting Area (Middle Left)**: Nearby washroom facilities ensure convenience for guests.
- **Storage:**

  Storage is integrated subtly, such as cabinets in the kitchen and potential storage space in the reception for supplies and materials.
  - **Kitchen Area (Top Left)**: May include cabinets for food, utensils, and cleaning supplies.
  - **Dining (Center)**: Could have space for storing additional dining supplies.
  - **Reception (Top Right)**: May include cabinets for visitor logs and administrative tools.
  - **Washroom Area (Bottom Right)**: Minimal storage, primarily for hygiene products.
  - **Waiting Area (Middle Left)**: Potential for magazines or décor storage.
- **Aesthetic Additions:**

  Plants and a well-furnished waiting area enhance the facility's appeal, creating a welcoming and pleasant environment.
  - **Kitchen Area (Top Left)**: A tidy layout and potential decorative elements like shelves enhance the space.
  - **Dining (Center)**: The arrangement of the table and chairs creates a professional yet welcoming environment.
  - **Reception (Top Right)**: A neat design with possibly decorative elements like signage or branding.
  - **Washroom Area (Bottom Right)**: Clean, simple design ensures user comfort.
  - **Waiting Area (Middle Left)**: Plants and furniture placement create a relaxed, inviting ambiance.
- **Furniture Distribution:**

  Furniture placement, like the organized dining setup and seating in the waiting area, balances functionality with accessibility, ensuring comfort and productivity.
  - **Kitchen Area (Top Left)**: Sink, counters, and cabinets are strategically placed for functionality.
  - **Dining (Center)**: The oval table and chairs are spaced to accommodate multiple users without feeling crowded.
  - **Reception (Top Right)**: A desk and seating for visitors ensure a functional layout.
  - **Washroom Area (Bottom Right)**: Fixtures are arranged to maximize utility while preserving privacy.
  - **Waiting Area (Middle Left)**: Couch and TV create a comfortable seating area for guests.

- **Piping System**:

  A piping system is a network of pipes and related components designed to transport fluids such as water, gas, or other substances within a facility. It is vital for ensuring access to resources like clean water and gas across different compartments.

- **Kitchen Area (Top Left)**: Piping provides water for the sink and possibly gas lines for cooking appliances.
- **Washroom Area (Bottom Right)**: Supplies water to toilets, sinks, and other hygiene fixtures.
- **Dining Area (Center)**: May include water dispensers or connections for drinking facilities.
- **Reception (Top Right)**: Minimal piping may exist for a nearby handwashing station, if required.
- **Pathways and Hallways**: Pipes are often routed through walls or beneath floors in these areas to connect different spaces seamlessly.

- **Sewage System**:
  A sewage system is an infrastructure that collects, transports, and disposes of wastewater and waste materials generated within a facility. It ensures cleanliness and prevents contamination.
  - **Kitchen Area (Top Left)**: Drain wastewater from sinks and other kitchen operations into the sewage system.
  - **Washroom Area (Bottom Right)**: Includes sewage pipes for managing toilet waste, sink water, and other hygiene-related drainage.
  - **Dining Area (Center)**: May have indirect drainage systems for water dispensers or accidental spills.
  - **Hallways and Pathways**: Sewage pipes run below the floors to transport waste to external disposal or treatment facilities.

4. Analysis and Discussion

1. Location Selection Analysis

- **Optimal Location**: Mandi Mor Rawalpindi emerged as the most suitable location, achieving the highest score due to its strategic **proximity to customers** and excellent **accessibility**, ensuring high foot traffic and ease of access for both customers and suppliers.
- **Supporting Factors**: While regulatory compliance and cost considerations were secondary, they contributed positively to the decision-making process, ensuring feasibility and alignment with budgetary constraints.

2. Layout Design Evaluation

- **Block Layout**: The proposed block layout prioritized the logical flow of operations, maintaining the separation of critical zones such as the kitchen, dining area, and restrooms to enhance efficiency and hygiene.
- **Detailed Layout**: The refined layout design focused on reducing **employee walking distances**, strategically positioning zones to support seamless workflows, and improving customer service delivery.

3. Optimization Results

- **Cost Reduction**: The application of the **CRAFT Method** and **Graph-Based Optimization Techniques** achieved a **15% reduction in operational costs**, significantly improving the cost-efficiency of the facility.
- **Workflow Improvements**: Simulations validated the optimized layout, demonstrating enhanced workflow efficiency, minimized bottlenecks, and improved customer convenience. The placement of zones based on adjacency priorities streamlined operations and reduced material handling efforts.

5. Conclusion

This project demonstrates the successful application of engineering tools and methodologies to design a high-performing restaurant facility. The following outcomes were achieved:

- **Optimal Location Selection**: The Analytic Hierarchy Process (AHP) identified Mandi Mor Rawalpindi as the most suitable location, ensuring proximity to customers and ease of accessibility.
- **Efficient Layout Design**: Factory CAD, along with CRAFT and graph-based optimization techniques, facilitated the development of a logical and efficient facility layout. This layout minimizes operational costs, reduces employee walking distances, and ensures smooth workflow.

Attributes of Complex Engineering Problem

Sr. No	Attribute	Explanation
1	Preamble	The project tackled the complex issue of designing a restaurant facility, requiring location selection, layout optimization, and validation through simulations.
2	Range of conflicting requirements	Balancing customer accessibility, operational costs, and compliance with local regulations required trade-offs to meet diverse and conflicting criteria.
3	Depth of analysis required	Extensive use of AHP for multi-criteria decision-making, layout optimization via Factory CAD, and simulation ensured comprehensive problem-solving.
4	Depth of knowledge required	Demanded expertise in industrial engineering principles, optimization techniques (CRAFT, graph-based), and CAD tools for facility layout design.
5	Familiarity of issues	Issues like layout optimization, cost management, and location decision-making are moderately familiar but required contextual adaptation for the restaurant.
6	Extent of applicable codes	Involved adherence to local zoning laws and safety standards while considering industrial engineering design principles.
7	Extent of stakeholder involvement and level of conflicting requirements	Included considerations of customer preferences, regulatory bodies, and cost constraints, necessitating stakeholder analysis and compromise on some criteria.
8	Consequences	Poor design could lead to inefficiency, customer dissatisfaction, and regulatory penalties; successful implementation ensured cost savings and operational ease.
9	Interdependence	The success of the project depended on integrating location selection, workflow design, and simulation validation for a coherent and efficient facility setup.

Assessment Rubrics

Knowledge Component	**Domain**	**PLO mapped**	**Taxonomy Level**	**Max Marks**	**Obtained Marks**
Each group should use AHP to select the appropriate location of facility (**Evaluate)**	Cognitive	2	C-4	5	
Apply Factory cad / linear programming to model the problem	Cognitive	1	C-2	5	
Analyze different parameters using Craft method or any other optimization method	Cognitive	2	C-3	5	
Analyze designing/ improvement of an industrial facility using graph based method	Cognitive	2	C-3	5	
Consider all other factors of facility design (**Analyze)**	Cognitive	2	C-3	5	
Create Design of facility	Cognitive	3	C-6	5	
Apply facility design principles to one's own work, as a member and /or leader in a team	Affective	11	A-2	5	
Manage facility design project in the broader context of innovation and technological developments.	Affective	11	A-2	5	